1875 ▸ 1885

1876
Invention of the telephone
American inventor Alexander Graham Bell patented the telephone. His device for transmitting and receiving human speech converted sound vibrations into electrical signals, and vice versa.

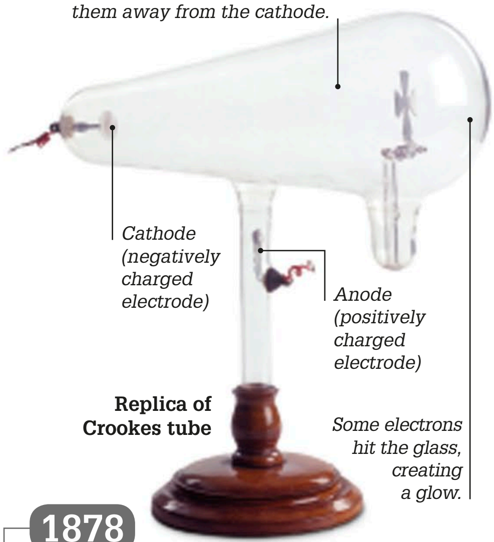
Communication
See pages 150–151

1876
Bacterial breakthrough
Robert Koch, a German bacteriologist, showed that anthrax—a deadly, infectious disease of sheep and cattle that can spread to humans—is transmitted by a bacterium called *Bacillus anthracis*. His finding confirmed French chemist Louis Pasteur's germ theory (see p.143).

The canals turned out to be optical illusions.

Artist's depiction of Schiaparelli's map of Mars

1877
Life on Mars
Italian astronomer Giovanni Schiaparelli reported seeing *canali* on Mars. *Canali* simply means "channels" in Italian, but it was mistakenly translated into English as "canals." This caused a storm of rumors that Mars was inhabited by intelligent beings.



Replica of Crookes tube

1878
Crookes tube
William Crookes, an English chemist, invented the cathode ray tube (also called Crookes tube). This tube, holding only a miniscule amount of gas, contained a negatively charged electrode (an electrical conductor) called a cathode, as well as a positively charged one, an anode. It would later be used to discover electrons and X-rays (see p.158).

1875

1876 FOUR-STROKE ENGINE

German inventor Nikolaus Otto invented the four-stroke internal combustion engine, which ignites fuel inside a cylinder. The Otto engine operates on a four-stroke cycle that moves a piston up and down in the cylinder. German engineer Karl Benz later adapted it as an automobile engine.

This model of an Otto internal combustion engine from 1886 burned gas and was stationary.